

Laplace Transformation

3. Properties of Laplace Transformation:

Laplace transform of derivatives:

$$L(f'(t)) = sL(f) - f(0)$$

$$L(f'') = s^2L(f) - sf(0) - f'(0)$$

$$L(f^n) = S^n L(f) - S^{n-1} f(0) - S^{n-2} f'(0) \dots S^0 f^{(n-1)}(0)$$

Using Laplace transform, we will convert the differential equation into an algebraic equation.

Question: A differential equation with boundary conditions is given as

$$y'' - y = t \text{ \& } y(0) = 1, y'(0) = 1. \text{ Then what will be its Laplace and inverse Laplace}$$

transform?

Solution: $L\{y'' - y\} = L\{t\}$

$$L\{y''\} - L\{y\} = \frac{1}{s^2}$$

$$s^2 L\{y\} - sy(0) - y'(0) - L\{y\} = \frac{1}{s^2}$$

$$s^2 L\{y\} - s - 1 - L\{y\} = \frac{1}{s^2}$$

$$(s^2 - 1)L\{y\} = \frac{1}{s^2} + s + 1 \Rightarrow L\{y\} = \frac{1}{(s^2 - 1)} \left(\frac{1}{s^2} + s + 1 \right)$$

$$L\{y\} = \frac{1}{(s^2 - 1)} \left[\frac{s^3 + s^2 + 1}{s^2} \right]$$

$$= \frac{s^3 + s^2 + 1}{(s-1)(s+1)s^2} = \frac{A}{(s-1)} + \frac{B}{(s+1)} + \frac{(Cs+D)}{s^2}$$

$$A = \frac{3}{2}, B = \frac{-1}{2}, C = 0 \text{ \& } D = -1$$

$$\text{Then } L\{y\} = \frac{s^3 + s^2 - 1}{(s+1)(s-1)s^2} = \frac{3}{2(s-1)} - \frac{1}{2(s+1)} - \frac{1}{s^2}$$

$$L^{-1}\{L(y)\} = L^{-1}\left\{ \frac{3}{2(s-1)} - \frac{1}{2(s+1)} - \frac{1}{s^2} \right\}$$

$$y = \frac{3}{2} L^{-1}\left\{ \frac{1}{s-1} \right\} - \frac{1}{2} L^{-1}\left\{ \frac{1}{s+1} \right\} - L^{-1}\left\{ \frac{1}{s^2} \right\}$$

$$y = \frac{3}{2} e^t - \frac{1}{2} e^{-t} - t = e^t + \frac{e^t}{2} - \frac{e^{-t}}{2} - t$$

$$y = e^t + \sinh t - t$$

Question: A differential equation is given by $y'' - y' - 6y = 0$ which satisfy the following boundary conditions $y(0) = 11$ & $y'(0) = 28$. So find its Laplace and inverse Laplace transform.

Solution: $L\{y'' - y' - 6y\} = 0 \Rightarrow L\{y''\} - L\{y'\} - 6L\{y\} = 0$

$$s^2 L\{y\} - sy(0) - y'(0) - sL\{y\} + y(0) - 6L\{y\} = 0$$

$$= (s^2 - s - 6)L\{y\} = sy(0) + y'(0) - y(0) = s(11) + 28 - 11$$

$$(s^2 - s - 6)L\{y\} = 11s + 17$$

$$L\{y\} = \frac{11s+17}{s^2-s-6} = \frac{11s+17}{(s-3)(s+2)}$$

$$\frac{11s+17}{(s-3)(s+2)} = \frac{A}{s-3} + \frac{B}{s+2}$$

$$A=10 \quad \& \quad B=1$$

$$\text{So, } L\{y\} = \frac{10}{(s-3)} + \frac{1}{(s+2)}$$

$$L^{-1}\{L\{y\}\} = 10L^{-1}\left\{\frac{1}{s-3}\right\} + L^{-1}\left\{\frac{1}{s+2}\right\} = 10e^{3t} + e^{-2t}$$

Question: $L^{-1} \frac{1}{s(s^2 + \omega^2)}$

Solution: $L^{-1} \frac{1}{(s^2 + \omega^2)} = \frac{\sin \omega t}{\omega}$

$$\text{Then, } L^{-1}\left\{\frac{1}{s(s^2 + \omega^2)}\right\} = \int_0^t \frac{\sin \omega t}{\omega} dt = \left[-\frac{\cos \omega t}{\omega^2}\right]_0^t = \frac{-1}{\omega^2}[\cos \omega t - 1]$$

$$L^{-1} \frac{1}{s(s^2 + \omega^2)} = \frac{1}{\omega^2}[1 - \cos \omega t]$$

Question: $L^{-1} \frac{1}{s^2(s^2 + \omega^2)}$

Solution: $L^{-1} \frac{1}{(s^2 + \omega^2)} = \frac{\sin \omega t}{\omega}$

$$\text{Then, } L^{-1}\left\{\frac{1}{s(s^2 + \omega^2)}\right\} = \int_0^t \frac{\sin \omega t}{\omega} dt = \frac{1}{\omega^2}[1 - \cos \omega t]$$

$$L^{-1}\left[\frac{1}{s}\left\{\frac{1}{s(s^2 + \omega^2)}\right\}\right] = \frac{1}{\omega^2} \int_0^t [1 - \cos \omega t] dt$$

$$L^{-1}\left[\frac{1}{s}\left\{\frac{1}{s(s^2 + \omega^2)}\right\}\right] = \frac{1}{\omega^2}\left[t - \frac{\sin \omega t}{\omega}\right]_0^t$$

$$L^{-1} \frac{1}{s^2(s^2 + \omega^2)} = \frac{1}{\omega^2}\left[t - \frac{\sin \omega t}{\omega}\right] = \frac{1}{\omega^3}[\omega t - \sin \omega t]$$

Question: $f(t) = \sin^2 \omega t$

Solution: $f(t) = \sin^2 \omega t = \frac{1 - \cos 2\omega t}{2}$

$$L\{f(t)\} = \frac{1}{2} L\{1 - \cos 2\omega t\} = \frac{1}{2} \left\{ \frac{1}{s} - \frac{s}{s^2 + 4\omega^2} \right\}$$

Or

$$f(t) = \sin^2 \omega t$$

$$f'(t) = 2\omega \sin \omega t \cos \omega t = \omega \sin 2\omega t$$

$$L\{f'(t)\} = sL\{f(t)\} - f(0)$$

$$\omega \cdot \frac{2\omega}{\omega^2 + 4\omega^2} = sL\{f(t)\} - 0$$

$$L\{f(t)\} = \frac{2\omega^2}{s(s^2 + 4\omega^2)}$$

Question: $\frac{dy}{dx} + ay = e^{-bt}$

$$y(0) = 0 \quad Y(s) = ?$$

Solution: $L\left\{\frac{dy}{dt} + ay\right\} = L\{e^{-bt}\}$

$$= sL\{y(t)\} + aL\{y\} = \frac{1}{s+b} = sy - y(0) + ay = \frac{1}{s+b}$$

$$y = \frac{1}{(s+b)(s+a)} \quad \{\because y(0) = 0\}$$